

1.ADF test with linear trend

- a. Recall from Assignment 1 the model selection process of an $AR(p)$ model with linear trend. In symbols (i.e. no need to put in numerical parameter estimates), write out the trend equation and autoregressive equation with selected imposed. Based on this, write out the implied ADF test equation in symbols (no derivation required). Write out the hypotheses for the resulting unit root test.

Trend residual eq

$$Z_t = \phi_1 Z_{t-1} + \phi_2 Z_{t-2} + \varepsilon_t$$

$$Y_t = \beta_0 + \beta_1 \text{Time}_t + \phi_1 Z_{t-1} + \phi_2 Z_{t-2} + \varepsilon_t$$

$$\Delta Y_t = \alpha_0 + \alpha_1 \text{Time}_t + \rho Y_{t-1} + \gamma_1 \Delta Y_{t-1} + \varepsilon_t$$

where $\rho = \phi_1 + \phi_2 - 1$ and $\gamma_1 = -\phi_2$ (one augmenting lag term since $p_0 = 2$).

$H_0 : \rho = 0$ (Y_t has a unit root — non-stationary, trend-stationary alternative not

$H_1 : \rho < 0$ (Y_t is trend-stationary — reject unit root)

- b. Use the `ur.df()` command to carry out the unit root test for the log gold price over the estimation sample with number of lags given in your answer in part (a). Compute the corresponding p -value and complete the hypothesis test. What is the implication of this test for modelling the log gold price?

```
> ADF <- ur.df(Y, type="trend", lags=1)
> print(round(ADF@testreg$coefficients,4))
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.0598      0.0471   1.2695  0.2053
z.lag.1      -0.0088      0.0081  -1.0793  0.2814
tt           0.0000      0.0001   0.5880  0.5570
z.diff.lag   0.1365      0.0604   2.2607  0.0246
>
> p_value <- punitroot(ADF@teststat[1], N = n, trend = "ct", statistic = "t")
> print(p_value)
[1] 0.9294376
```

P value > 0.05, fail to reject the unit root, thus further differencing is required to model the log gold price by addressing the stochastic unit root trend, but does not address deterministic trend shifts such as a break.

- c. Show how the ADF test regression in part (b) can be replicated using a regression estimated with `lm()`. In particular, as required construct variables that enter the ADF test equation, and show that the same estimated ADF test statistics can be obtained using `lm()` and `ur.df()`.

```

> # p: lag length in ADF test equation
> p <- 1
>
> # observation indices for AR(p) model
> t <- (p+2):n # n + 2 for augmented lag
>
> # First difference and one lag of Y:
> DY <- Y[t]-Y[t-1]
> Y1 <- Y[t-1]
> DY1 <- Y[t-1]-Y[t-2]
>
> # ADF testing equation with linear trend
> eq0 <- summary(lm(DY ~ Y1 + t + DY1))
> print(round(eq0$coefficients, 4))

```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.0607	0.0480	1.2640	0.2074
Y1	-0.0090	0.0083	-1.0800	0.2811
t	0.0000	0.0001	0.6324	0.5277
DY1	0.1363	0.0616	2.2123	0.0278

- d. Using the conclusion to the unit root test, write out the implied model for the log gold price, in symbols, derived from the model in part (a).

Since the ADF test failed to reject $H_0: \rho = 0$, the unit root restriction $\phi_1 + \phi_2 = 1$ is imposed on the AR(2)-with-trend model from part (a):

$$Y_t = \beta_0 + \beta_1 \text{Time}_t + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \varepsilon_t$$

Substituting $\phi_2 = 1 - \phi_1$ and rearranging in first-difference form gives the **implied model**:

$$\Delta Y_t = \beta_0 + \beta_1 \text{Time}_t + \gamma_1 \Delta Y_{t-1} + \varepsilon_t$$

where $\gamma_1 = \phi_1 - 1$ (equivalently $\gamma_1 = -\phi_2$).

- e. Estimate the model from (d) over the estimation sample, report the coefficient estimates, and write out the estimated model in equation form.

Estimated coefficients (over the estimation sample, n = 262 usable observations after differencing/lagging):

	Estimate	Std. Error	t value	Pr(> t)
Intercept ($\hat{\beta}_0$)	0.5801	0.7143	0.8122	0.4174
Time ($\hat{\beta}_1$)	-0.0003	0.0004	-0.8040	0.4221
ΔY_{t-1} ($\hat{\gamma}_1$)	0.1314	0.0615	2.1367	0.0336

$R^2 = 0.0206$

Estimated model in equation form:

$$\Delta \hat{Y}_t = 0.5801 - 0.0003 \cdot \text{Time}_t + 0.1314 \cdot \Delta Y_{t-1}$$

- a. Recall the autoregressive lag length and trend break choices from Assignment 1. Write out the trend equation and autoregressive equation in symbols. Based on this, write out the implied ADF test equation in symbols (no derivation required). Write out the hypotheses for the resulting unit root test.

Lag = 3

- Selected break date: $\hat{b} = 2011.417$ (minimises RSS over the candidate range Jan 2010 – Dec 2014).

where $TB_t(\hat{b}) = 1(\text{Time}_t > \hat{b}) \times (\text{Time}_t - \hat{b})$, and $\hat{b} = 2011.417$ (June 2011).

$$Y_t = \alpha_0 + \alpha_1 \text{Time}_t + \alpha_2 TB_t(\hat{b}) + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \phi_3 Y_{t-3} + \varepsilon_t$$

$$\Delta Y_t = \alpha_0 + \alpha_1 \text{Time}_t + \alpha_2 TB_t(\hat{b}) + \rho Y_{t-1} + \gamma_1 \Delta Y_{t-1} + \gamma_2 \Delta Y_{t-2} + \varepsilon_t$$

where $\rho = \phi_1 + \phi_2 + \phi_3 - 1$, $\gamma_1 = -(\phi_2 + \phi_3)$, and $\gamma_2 = -\phi_3$.

Hypotheses (unit root test):

$H_0 : \rho = 0$ (Y_t has a unit root — non-stationary, even after allowing for a deterministic

$H_1 : \rho < 0$ (Y_t is trend-stationary around the broken trend — reject unit root)

- b. Note that the `ur.df()` command does not have an option to include a trend break in the test equation along with intercept and trend. Therefore, based on your answer in (a), set up the ADF test equation for estimation with `lm()`. Report the coefficient estimates and t statistics in table form, and write out the estimated model in equation form. Give the value of the ADF t statistic.

Estimated Model in Equation Form

$$= -14.8948 + 0.0076 \text{Time}_t - 0.0076 TB_t(\hat{b}) - 0.0478 Y_{t-1} + 0.1658 \Delta Y_{t-1} - 0.0686 \Delta Y_{t-2}$$

```

Call:
lm(formula = DY ~ Y1 + tt + TBt + DY1 + DY2)

Residuals:
    Min       1Q   Median       3Q      Max
-0.137225 -0.020817 -0.002122  0.020319  0.108537

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -14.894837    5.800363  -2.568  0.01080 *
Y1           -0.047778    0.017527  -2.726  0.00685 **
tt            0.007580    0.002945   2.574  0.01062 *
TBt          -0.007645    0.002999  -2.549  0.01139 *
DY1           0.165806    0.061421   2.700  0.00741 **
DY2          -0.068600    0.061630  -1.113  0.26672
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.03556 on 255 degrees of freedom
Multiple R-squared:  0.05996,    Adjusted R-squared:  0.04153
F-statistic: 3.253 on 5 and 255 DF,  p-value: 0.007255

```

Coefficient Estimates

	Estimate	Std. Error	t value	Pr(> t)
Intercept ($\hat{\alpha}_0$)	-14.8948	5.8004	-2.5679	0.0108
Y_1 ($\hat{\rho}$)	-0.0478	0.0175	-2.7260	0.0069
tt ($\hat{\alpha}_1$)	0.0076	0.0029	2.5741	0.0106
TBt ($\hat{\alpha}_2$)	-0.0076	0.0030	-2.5491	0.0114
$DY1$ ($\hat{\gamma}_1$)	0.1658	0.0614	2.6995	0.0074
$DY2$ ($\hat{\gamma}_2$)	-0.0686	0.0616	-1.1131	0.2667

ADF Test Statistic

$$\tau_3 = -2.7260$$

- c. Set up a simulation to obtain the appropriate 5% critical value for the ADF t test with trend and trend break in part (b) to apply to the unit root test on the log gold price time series.

```

> cv_5 <- quantile(tADF, 0.05)
> print(cv_5 )
      5%
-3.852213

```

- d. Use the results of your simulation to compute the p -value for the ADF t statistic from (c). Complete the unit root test and explain the implication of the conclusion for modelling the log gold price.

```
> tau_obs <- -2.7260 # observed ADF stat
> p_value_sim <- mean(tADF <= tau_obs)
> print(p_value_sim)
[1] 0.476
```

Implication for Modelling the Log Gold Price

The conclusion is unchanged from the simpler trend-only ADF test, and in fact even more decisively so once the correct (simulated) critical values are used: even after allowing for both a deterministic linear trend **and** a structural break in that trend at June 2011, the log gold price series still exhibits a unit root — it is non-stationary in levels.

This has several concrete modelling implications:

1. **The levels-based trend/trend-break models from Q2-Q3 are misspecified for inference purposes.** Regressing Y_t on a deterministic trend (with or without a break) and AR terms, when Y_t is actually $I(1)$, risks spurious regression problems — high R^2 and seemingly significant trend/break coefficients can arise mechanically from the persistence of an integrated series, rather than reflecting genuine deterministic structure. This explains why the earlier trend-break model achieved a strikingly high R^2 (0.9573) despite the series fundamentally being a unit-root process — that fit is partly an artefact of fitting a flexible deterministic function to a stochastically trending series, not evidence the trend-break specification is "correctly" capturing the data-generating process.
2. **The appropriate remedy is differencing, not a better-fitting deterministic trend.** Since the non-stationarity is stochastic (unit root) rather than purely deterministic, adding more deterministic structure (trends, breaks, dummies) cannot fix the underlying issue. The series should instead be modelled in first differences, ΔY_t , which — as shown in the earlier estimated $AR(1)$ -in-differences model — is consistent with the unit root restriction and avoids the spurious regression concern.
3. **Forecasting implications:** despite this, the Q4 recursive forecasting exercise showed the $AR(p)$ +trend and $AR(p)$ +trend+break models still performed reasonably well for *short-horizon forecasting* ($RMSE \approx 0.035$). This is a known feature of near-unit-root series — trend-stationary models can still produce usable one-step-ahead forecasts even when they're not the "correct" data-generating process, because a well-fitted deterministic trend can locally approximate a slowly evolving stochastic trend over short forecast horizons. However, this masks the deeper structural issue: for longer-horizon forecasting, or for any structural/inferential use of the estimated coefficients (e.g. testing significance of the trend break, or computing valid confidence intervals), the levels-based models remain unreliable, and a differenced ($I(1)$) specification such as $ARIMA(p,1,q)$ would be the econometrically defensible choice going forward.

Completing the Unit Root Test

	Value
Observed ADF statistic (τ_y)	-2.7260
Simulated 5% critical value	-3.9456
Simulated p-value	0.443

Decision rule: reject H_0 if $\tau_y < \text{critical value}$ (i.e. more negative), or equivalently if $p\text{-value} < 0.05$.

- -2.7260 is **not** more negative than -3.9456
- 0.443 is **far greater** than 0.05

→ Fail to reject H_0 : $\rho = 0$.

- e. Derive the form of the model, in symbols, for the log gold price implied by the conclusion in (d). Specifically, derive the forms of the trend equation and autoregressive equation that would apply.

Since the ADF test with trend and break also failed to reject $H_0: \rho = 0$, the unit root restriction $\phi_1 + \phi_2 + \phi_3 = 1$ is imposed on the AR(3)-with-trend-and-break model.

Starting Point — Trend Equation and AR Equation (from earlier)

Trend equation with break:

$$Y_t = \beta_0 + \beta_1 \text{Time}_t + \beta_2 \text{TB}_t(\hat{b}) + Z_t$$

AR(3) equation for the residual:

$$Z_t = \phi_1 Z_{t-1} + \phi_2 Z_{t-2} + \phi_3 Z_{t-3} + \varepsilon_t$$

Combined, in levels of Y:

$$Y_t = \alpha_0 + \alpha_1 \text{Time}_t + \alpha_2 \text{TB}_t(\hat{b}) + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \phi_3 Y_{t-3} + \varepsilon_t$$

Imposing the Unit Root Restriction

The unit root null is $\phi_1 + \phi_2 + \phi_3 = 1$. Substituting this restriction in (equivalently, setting $\rho = \phi_1 + \phi_2 + \phi_3 - 1 = 0$ in the ADF reparametrisation derived earlier) removes the Y_{t-1} level term entirely, leaving only the augmenting difference terms and the deterministic components:

$$\Delta Y_t = \alpha_0 + \alpha_1 \text{Time}_t + \alpha_2 \text{TB}_t(\hat{b}) + \gamma_1 \Delta Y_{t-1} + \gamma_2 \Delta Y_{t-2} + \varepsilon_t$$

where $\gamma_1 = -(\phi_2 + \phi_3)$ and $\gamma_2 = -\phi_3$, exactly as derived previously.

Implied Model — Trend Equation and Autoregressive Equation

Splitting this back into its two components (matching the structure of your original question):

Trend equation (now on the differenced series):

$$\Delta Y_t = \beta_0 + \beta_1 \text{Time}_t + \beta_2 \text{TB}_t(\hat{b}) + Z_t^\Delta$$

Autoregressive equation for the residual, AR(2) in the difference:

$$Z_t^\Delta = \gamma_1 Z_{t-1}^\Delta + \gamma_2 Z_{t-2}^\Delta + \varepsilon_t$$

Combined:

$$\Delta Y_t = \beta_0 + \beta_1 \text{Time}_t + \beta_2 \text{TB}_t(\hat{b}) + \gamma_1 \Delta Y_{t-1} + \gamma_2 \Delta Y_{t-2} + \varepsilon_t$$

Estimated Model in Equation Form

$$\Delta \hat{Y}_t = 0.3666 - 0.0002 \text{Time}_t - 0.0004 \text{TB}_t(\hat{b}) + 0.1508 \Delta Y_{t-1} - 0.0916 \Delta Y_{t-2}$$